

June 17, 2026

Dear Admission Committee,

I am writing to express my strong interest in pursuing a PhD in Quantum Technology Engineering at the University of Southampton, located in Southampton, United Kingdom. The project on "Sparse Variational Quantum Machine Learning" particularly resonates with me, given my background and expertise in AI Data Science.

As I delve into the details of this project, I am excited to see how the proposed adaptation of methods from sparse optimization can revolutionize the field of variational quantum algorithms (VQAs). My experience with developing a linear-time state space model for sub-millisecond bedside inference, *MedicalMambaTS*, has given me a unique perspective on the importance of latency and computational efficiency.

Furthermore, I am impressed by the potential of quantum computers to solve problems that cannot be efficiently solved on traditional computers, as well as the limitations imposed by system noise and decoherence. As someone who has worked with physical constraints in a 0D Windkessel Physics-Informed Operator Network using PyTorch Autograd (*PI_DeepONet*), I am eager to explore how these concepts can be applied to optimize quantum systems.

I am confident that my background and expertise, combined with the University of Southampton's research excellence, would make me an excellent candidate for this PhD program. I am excited about the opportunity to work on this project and contribute to the advancement of knowledge in the field of Quantum Technology Engineering.

Sincerely,

Ismail Aïssaoui